Exam practice: inside the CPU

Real OCR GCSE Computer Science questions. Write your answer first. Then mark it against the mark scheme on the board.

Name: _____ Total: _____ / 5

Q1 A table gives this description of a CPU component: 'Performs mathematical calculations and logical operations.' Write the name of the component. **[1]**

June 2022 Q2 (one row of a 4-mark table)

Mark scheme: Arithmetic Logic Unit // ALU. My mark: _____ / 1

Q2 Write the definition of the CU (Control Unit). **[1]**

June 2022 Q2 (one row of a 4-mark table)

Mark scheme: (Sends signals to) synchronise / control / coordinates the processor / hardware / F-E cycle / processes / flow of data // decodes instructions (in CIR) // runs F-E cycle. Guidance: CU 'sends signals to components' is not enough, it isn't saying what the signal's purpose is. Examiner report: saying the Control Unit actually performs the cycle or executes the instructions is a misconception. My mark: _____ / 1

Q3 Which component stores frequently used instructions? Choose from: MAR, MDR, Cache, Program Counter, RAM. **[1]**

OCR J277 question from the PMT 1.1 topic collection, Q5(a)(i) (one row of a 4-mark tick table)

Mark scheme: Cache. My mark: _____ / 1

Q4 A computer has a Central Processing Unit (CPU). Describe what happens during the fetch-execute cycle. **[2]**

June 2024 Q6(a)

Mark scheme: 1 mark each, any two: data/instructions are fetched from memory/RAM/primary storage; data/instructions are stored using the registers (or a correct example of a register storing an address or data); data/instructions are decoded (or split into opcode and operand); data/instructions are executed/processed; ALU performs the logical/arithmetic calculations. Ignore inaccurate references to registers and components. Examiner report: some said 'information was processed' or that 'programs were fetched from memory'. My mark: _____ / 2

Common wrong answers (examiner reports). Q1: none recorded. Q2: 'sends signals to components' alone earns nothing; saying the Control Unit executes the instructions is a misconception. Q3: none recorded. Q4: some candidates said information was processed, or that programs were fetched from memory (it is one instruction at a time).